

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester V Examination, 2016-17

Paper: BCS-502 — Operating System Concepts

Time: Three Hours

Maximum Marks: 70

Instructions: Answer five questions in all, including Question No.1, which is compulsory. Figures or 8 Pets

Question 1. Answer any six of the following parts: [14]

- (a) Define the term Operating System and distinguish between the internal and external views of an operating system.
- (b) What are the main design goals of an operating system? Explain them in brief.
- (c) What is the starvation problem? Which CPU scheduling algorithms suffer from this problem?
- (d) Distinguish between a process and a thread. State the major advantages of using threads.
- (e) What is the System Resource-Allocation Graph? Describe.
- (f) When are Dynamic Loading and Dynamic Linking required?
- (g) Briefly discuss any three free-space management schemes for a disk.

Question 2.

- (a) Briefly explain the different process states and transitions among them with a state transition diagram. [2]
- (b) Explain the three-level scheduling that may coexist in a complex operating system. [4]
- (c) Consider the following five processes, with arrival times and CPU burst times given in milliseconds: [8]

Process	Arrival Time	Burst Time
P1	0	10
P2	2	29
P3	5	3
P4	8	7
P5	10	12

Illustrate the First-Come-First-Serve (FCFS), Non-Preemptive Shortest-Job-First (SJF), and Round-Robin (RR, with a time quantum of 10 milliseconds) scheduling algorithms using a Gantt chart. Which algorithm will give the minimum average waiting time?

Question 3.

- (a) What do you mean by memory fragmentation? Differentiate between internal and external memory fragmentation. [6]

- (b) Define the concepts of Logical and Physical Addresses. How is a logical address translated to a physical address in a paging scheme? Explain with a neat diagram. [8]

Question 4.

- (a) What is Demand Paging? Explain the steps involved in page fault handling. [4]
- (b) What is Belady's anomaly? Illustrate this anomalous behavior in the FIFO page replacement algorithm with a suitable example. [4]
- (c) Explain the Working Set Model. Explain how it is used to prevent thrashing and manage memory bandwidth in practice. [6]

Question 5.

- (a) Describe the physical and logical views of a disk storage system. [5]
- (b) Suppose that a disk drive has 5,000 cylinders, numbered 0 to 4999. The head is currently serving a request at cylinder 143, and the previous request was at cylinder 125. The queue of pending requests in FIFO order is:

86, 1470, 913, 1774, 948, 1509, 1622, 1756, 130

Starting from the current head position, what is the total distance (in cylinders) that the disk arm moves to satisfy all the pending requests for FCFS, SSTF, and SCAN disk scheduling algorithms? [9]

Question 6.

- (a) Describe the deadlock detection algorithm for a system with multiple instances of resource types. [6]
- (b) Consider the following snapshot of a system at time T_0 with 5 processes (P0–P4) and 4 resource types (A, B, C, D): [8]

Process	Allocation				Maximum				Available			
	A	B	C	D	A	B	C	D	A	B	C	D
P0	0	0	1	2	0	0	1	2	1	5	2	0
P1	1	0	0	0	1	7	5	0				
P2	1	3	5	4	2	3	5	6				
P3	0	6	3	2	0	6	5	2				
P4	0	0	1	4	0	6	5	6				

Answer the following questions using the Banker's algorithm:

- (i) What is the content of the matrix Need?
- (ii) Is the system in a safe state? If yes, give the safe sequence.
- (iii) If a request from process P1 arrives for (0, 4, 2, 0), can the request be granted immediately?

Question 7. Write short notes on the following:

[3 + 4 + 4 + 3 = 14]

- (a) Dining Philosophers Problem

- (b) Critical Section and Semaphores
- (c) Indexed File Allocation Method
- (d) Directory Structure